

PROBLEM STATEMENT

Power & Trigonometric Chain Rule

Find the derivative with respect to x of the function:

$$y = \sin^2(x^2)$$

STEP 1 প্রদত্ত ত্রিকোণমিতিক বর্গ ফাংশনটিকে বন্ধনী আকারে প্রকাশ করি (Expressing in Explicit Power Form)

$$y = [\sin(x^2)]^2$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ অপারেটর প্রয়োগ করি (Applying Differentiation Operator w.r.t. x)

$$\frac{dy}{dx} = \frac{d}{dx} ([\sin(x^2)]^2)$$

STEP 3 বহিঃস্থ ঘাত ২-এর জন্য শক্তি বিধি (Power Rule) প্রয়োগ করি (Applying Power Rule to Outer Exponent)

$$\frac{dy}{dx} = 2 [\sin(x^2)]^{2-1} \cdot \frac{d}{dx} [\sin(x^2)] = 2 \sin(x^2) \cdot \frac{d}{dx} [\sin(x^2)]$$

STEP 4 সাইন ফাংশন ও অভ্যন্তরীণ কোণ x^2 -এর অন্তরজ নির্ণয় করি (Differentiating Sine & Quadratic Argument)

$$\frac{d}{dx} [\sin(x^2)] = \cos(x^2) \cdot \frac{d}{dx}(x^2) = \cos(x^2) \cdot (2x) = 2x \cos(x^2)$$

STEP 5 গুণফল একত্রিত করে দ্বিকোণ সূত্রের মাধ্যমে সরল করি (Combining Terms & Applying Double-Angle Identity)

$$\frac{dy}{dx} = 2 \sin(x^2) \cdot [2x \cos(x^2)] = 4x \sin(x^2) \cos(x^2) = 2x \cdot [2 \sin(x^2) \cos(x^2)] = 2x \sin(2x^2)$$

FINAL ANSWER
$$\frac{dy}{dx} = 4x \sin(x^2) \cos(x^2) \quad (\text{or, } 2x \sin(2x^2))$$

BOX 1: MATHEMATICAL FOUNDATIONS & FORMULA REFERENCE

- Three-Tier Chain Rule:** Function structure $y = f(g(h(x)))$ where $f(u) = u^2$, $g(v) = \sin v$, and $h(x) = x^2$.
$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx} = (2u) \cdot (\cos v) \cdot (2x)$$
- Trigonometric Double-Angle Identity:** $\sin(2\theta) = 2 \sin \theta \cos \theta$.
- Power Rule:** $\frac{d}{dx}[u^n] = nu^{n-1} \frac{du}{dx}$.

BOX 2: COMMON STUDENT TRAPS & PITFALLS

- Notation Confusion:** $\sin^2(x^2) = [\sin(x^2)]^2 \neq \sin(x^4)$. Squaring the trig function is not the same as squaring the argument twice.
- Missing Inner Angle Derivative:** Forgetting to multiply by $\frac{d}{dx}(x^2) = 2x$, leaving the answer incorrectly as $2 \sin(x^2) \cos(x^2)$.
- Argument Squaring vs. Angle Multiplication:** Notice $\sin(2x^2) \neq \sin((2x)^2) = \sin(4x^2)$. The coefficient 2 multiplies the entire angle x^2 .

BOX 3: REAL DOMAIN & DIFFERENTIABILITY ANALYSIS

- Domain of Definition:** The sine function and polynomial x^2 are defined on the entire real line:
$$\text{Domain} = (-\infty, \infty) = \mathbb{R}$$
- Range:** Since $\sin(x^2) \in [-1, 1]$, $y = \sin^2(x^2) \in [0, 1]$.
- Differentiability:** Infinitely smooth (C^∞) everywhere on $\mathbb{R}$.

BOX 4: ALTERNATIVE METHOD VIA POWER-REDUCTION FORMULA

- Using identity $\sin^2 \theta = \frac{1 - \cos(2\theta)}{2}$:
$$y = \frac{1}{2} - \frac{1}{2} \cos(2x^2)$$
- Differentiating directly:
$$\frac{dy}{dx} = 0 - \frac{1}{2} (-\sin(2x^2)) \cdot \frac{d}{dx}(2x^2) = \frac{1}{2} \sin(2x^2) \cdot (4x) = 2x \sin(2x^2)$$
- This bypasses the power rule and provides the simplified double-angle form in one direct differentiation step!

BOX 5: EXAM MARKS BREAKDOWN (AMIE SECTION-A FORMAT)

- Step 1-2 (Power Formulation & Setup):** 1.0 Mark
- Step 3 (Power Rule Differentiation):** 1.0 Mark
- Step 4 (Inner Sine & Angle Differentiation):** 1.0 Mark
- Step 5 & Final Answer:** 1.0 Mark
- Total Allocation:** 4.0 Marks